Control unit operation

Wednesday, 29 March, 2023 11:45 AM

C_5

t_1
t_2 → P'Q' — Fetch — 00
t_3 P'Q — Indirect — 01
PQ' — Execute — 10
PQ = Interrupt — 11
IEC
PQ


control word
↓ memory

Example given in ch-12

Time sequence diagram

Single clock pulse
T-state

Memory read

MID



$\frac{128}{8} = 16$

$16M = 2^4 \times 2^{20}$
$= 2^{24} = 2^{12} \times 2^{12}$

$m = 12$
$n = 12$

Find:
b) Address bits
→ 12 bit.

Ans) 16 x 8

Modular dRAM

2.

$T_{without} = t \times b + [p \times (r \times s - 1) \times b]$

$T_{with} = t \times b + [p \times (r \times s - 1) \times d]$

$Speedup = \frac{T_{without}}{T_{with}}$

d) = 10

Types of clouds

Private :
Public :
Hybrid :

6, 7, 8,
12, 13, 14, ... 20)